

Classic Style

Theorem 1 | In a right triangle with legs a and b and hypotenuse c :

Pythagorean

$$a^2 + b^2 = c^2$$

Definition 1 | A **continuous function** is one where small changes in input produce small changes in output.

Lemma 1 | If f is continuous at a , then f is bounded in some neighborhood of a .

Remark | This result generalizes to higher dimensions.

Example 1 | Consider $f(x) = x^2$. Then $f(2) = 4$.

Proof. By direct calculation and the definition of continuity.

□